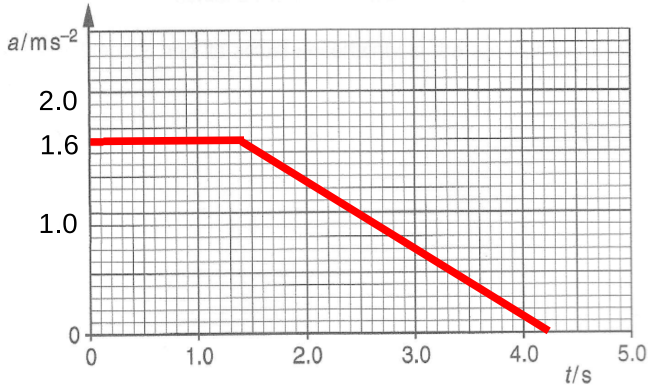
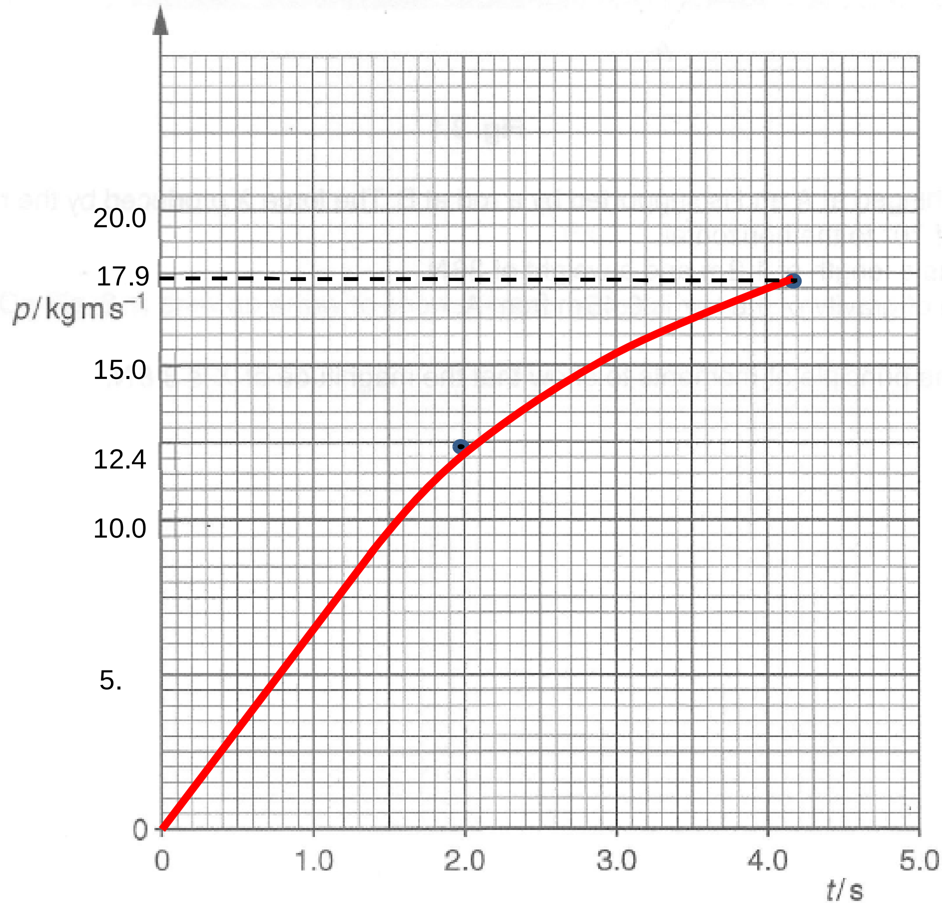
2 (a)	From Newton's 2nd law, i.e. $F = ma \Rightarrow a = \frac{F}{m} = \frac{6.4}{4.0} = 1.6 \text{ m s}^{-2}$

2(b)	
2(c)	Impulse $= \Delta p = F\Delta t = (6.4)(1.4) = 8.96$ N s
2(d)	For 0 to 4.2 s, Impulse = area under the graph $= \frac{1}{2}(6.4)(1.4 + 4.2) = 17.9$ N s  A significant number of answers suggested the momentum decreased in this time from 1.4 s to 4.2 s although a force is still being applied. Some answers gave an unsuitable momentum scale for the y-axis. There was insufficient accuracy in the drawing of the latter stage of the graph.

Comments for Question 2

(a) A minority of answers gave the force from the graph as 6.2 N or 6.0 N; other answers were correct.

(b) This was answered well.

(c) Some answers gave an impulse calculated for the entire duration of the force.

(d) The majority of answers applied the previous answer in **(c)** correctly to the graph for the first 1.4 s. A minority of answers gave the correct graph for the time from 1.4 s to 4.2 s.

A significant number of answers suggested the momentum decreased in this time although a force was still being applied. Some answers gave an unsuitable momentum scale for the y-axis. There was insufficient accuracy in the drawing of the latter stage of the graph.

3 (a) Taking moments about A,

$X \cos 40^\circ (1.2) = 36 \cos 60^\circ (0.45)$